

SOC Internship - Day 4

Topic: Phishing Attack, Malware Attack, and Brute Force Attack

1. Phishing Attack

A phishing attack is a type of social engineering attack. In this attack, the attacker does not first break the system technically. Instead, the attacker manipulates a person into giving access or sensitive information.

Attackers usually try to steal:

- Usernames and passwords
- Bank account details
- Debit or credit card information
- OTP codes
- Personal information
- Company login credentials

How Phishing Works

The attacker pretends to be a trusted source, such as:

- A bank
- Google or Microsoft
- A delivery company
- A government office
- A company HR team
- A social media platform

They send a message that creates fear, urgency, or curiosity.

Examples:

- “Your account will be suspended today.”
- “You have won a prize.”
- “Unusual login detected. Verify now.”
- “Your package delivery failed. Update address.”
- “Invoice attached. Please review immediately.”

When the victim clicks the link, downloads the file, or replies with details, the attacker may get access to the account or system.

Common Types of Phishing

1. Email phishing - A fake email looks like it came from a real company. Example: An email says your Gmail storage is full and asks you to log in through a link. The link opens a fake login page.
2. Spear phishing - A targeted phishing attack. The attacker uses the victim’s name, company, job role, or personal details to make the message believable.

3. Whaling - A phishing attack aimed at high-level people such as CEOs, managers, or directors.
4. Vishing - Phishing through voice calls. Example: Someone calls pretending to be from the bank and asks for an OTP or card details.
5. Clone phishing - A real email is copied, but the attachment or link is changed to a malicious one.

What Happens After a Phishing Attack

If a victim falls for phishing, the attacker may:

- Steal login credentials
- Access email or social media accounts
- Transfer money
- Install malware
- Steal company data
- Use the stolen account for more attacks

In companies, phishing can lead to:

- Data breaches
- Ransomware attacks
- Financial loss
- Reputation damage

2. Malware Attack

A malware attack is a cyberattack in which malicious software is used to damage, disrupt, spy on, or gain unauthorized access to a computer, phone, or network.

A malware attack can:

- Steal passwords and personal data
- Damage files
- Slow down the system
- Lock your device
- Spy on your activity
- Give attackers control of your computer
- Spread to other systems

Common Types of Malware

1. Virus - Attaches itself to a file or program. When the file is opened, the virus spreads and can damage data.
2. Worm - Spreads automatically from one computer to another through a network. It does not need user action like opening a file.
3. Trojan Horse - Looks like a useful or normal program, but it is actually harmful. Once installed, it can steal data or open backdoor access.
4. Ransomware - Locks or encrypts files and asks for money to unlock them.

5. Spyware - Secretly watches user activity and collects information such as passwords or browsing history.
6. Adware - Shows unwanted advertisements and may also track user behavior.
7. Keylogger - Records everything typed on the keyboard, including passwords and messages.
8. Rootkit - Hides inside the system and helps attackers maintain secret access.

How Malware Enters a System

- Infected email attachments
- Fake software downloads
- Phishing links
- Unsafe websites
- USB devices
- Pirated software
- Infected apps
- Network vulnerabilities

Signs of a Malware Attack

- Computer becomes slow
- Unknown pop-up ads appear
- Apps crash often
- Files disappear or become encrypted
- Unusual internet usage
- Browser redirects to strange websites
- Antivirus gets disabled
- Unknown programs appear

3. Brute Force Attack

A brute force attack is a cyberattack in which an attacker tries many passwords or combinations again and again until the correct one is found.

How It Works

The attacker uses software or scripts to try:

- Many passwords
- Number combinations
- Username and password pairs
- Encryption keys

The goal is to break into:

- Email accounts
- Social media accounts
- Bank accounts

- Websites
- Admin panels
- Wi-Fi passwords

Types of Brute Force Attacks

1. Simple brute force attack - The attacker manually or automatically tries many possible passwords until one works.
2. Dictionary attack - The attacker tries a list of common words or passwords.
3. Credential stuffing - The attacker uses usernames and passwords leaked from one website to try logging in on other websites.
4. Hybrid brute force attack - This combines common words with numbers or symbols.
5. Reverse brute force attack - The attacker starts with one common password and tries it on many usernames.

Why Brute Force Attacks Are Dangerous

- They can break into accounts
- They can steal personal data
- They can access bank details
- They can control websites
- They can spread malware
- They can cause financial loss

Signs of a Brute Force Attack

- Many failed login attempts
- Account lockouts
- Unusual login activity
- Login attempts from unknown IP addresses
- Sudden server slowdown

4. Assignment: Fake Bank Email Phishing Attack

Example Email

Subject: ⚠ Urgent: Your Bank Account Will Be Suspended

From: support@secure-hdfc.com

Dear Customer, We detected unusual activity on your account. Please verify your identity immediately to avoid account suspension. Click here to verify: <http://secure-hdfc-login-verification.com> Failure to verify within 24 hours will permanently lock your account.

Suspicious Signs

1. Fake domain name - The link is not the official bank website.

2. Urgent threats - Words like “account suspended”, “act immediately”, and “24 hours remaining” create pressure.

3. Generic greeting - Instead of using your real name, it says “Dear Customer”, “User”, or “Account Holder”.

How to Prevent Phishing

- Check the sender email carefully.
- Never click suspicious links.
- Enable two-factor authentication.
- Verify website URLs before entering login details.
- Avoid downloading unknown attachments.
- Use spam filters and antivirus software.